

DIGITAL LOGIC AND DESIGN

A COURSE INTRODUCTION

SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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ABOUT MYSELF



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- **Office Hours:**
 - TBD

WHAT IS A DIGITAL SYSTEM?

- Ability to represent and manipulate discrete elements of information.

Aircrafts guidance



Consumer Appliances



Heavy Industry



Medical

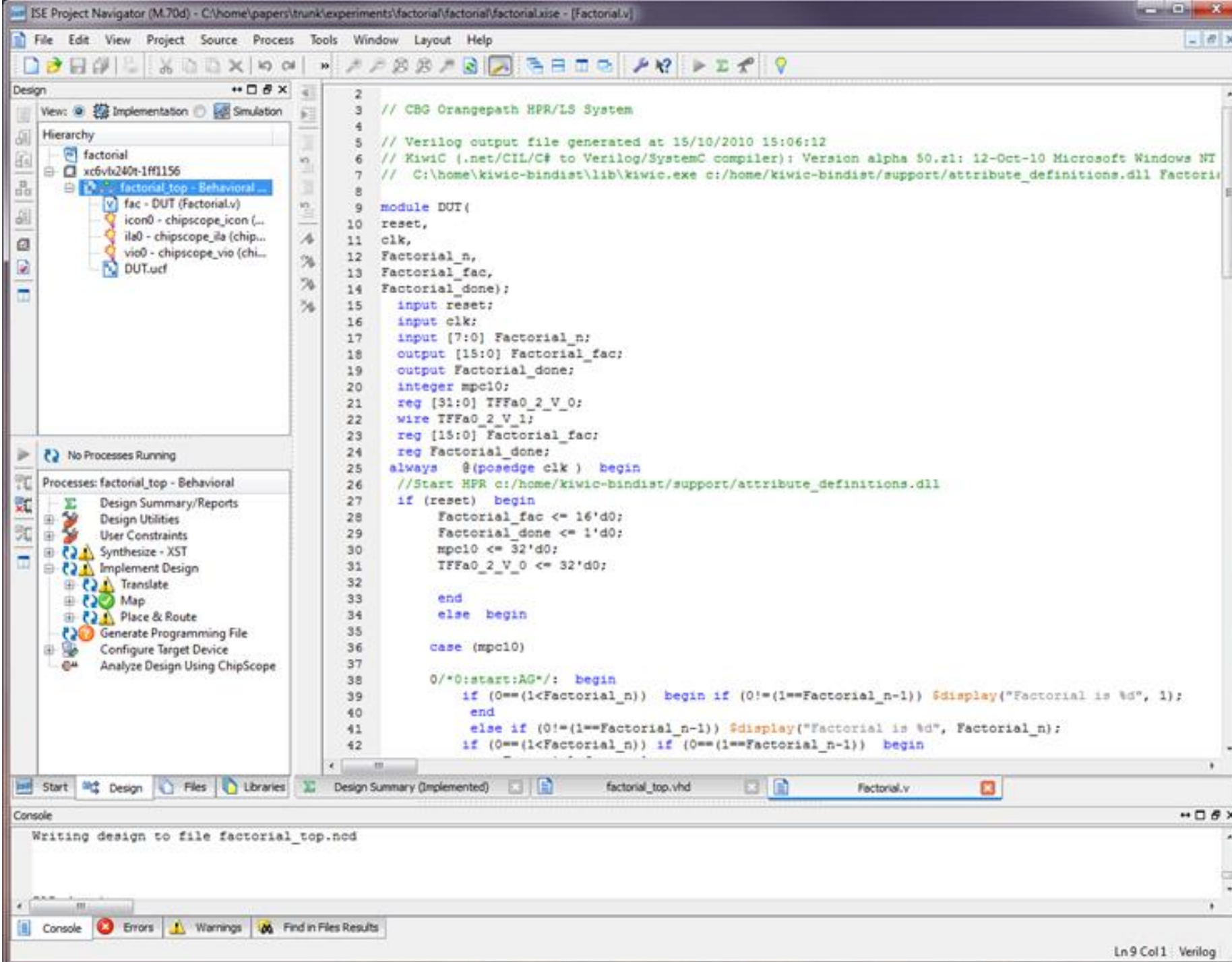


DNA research



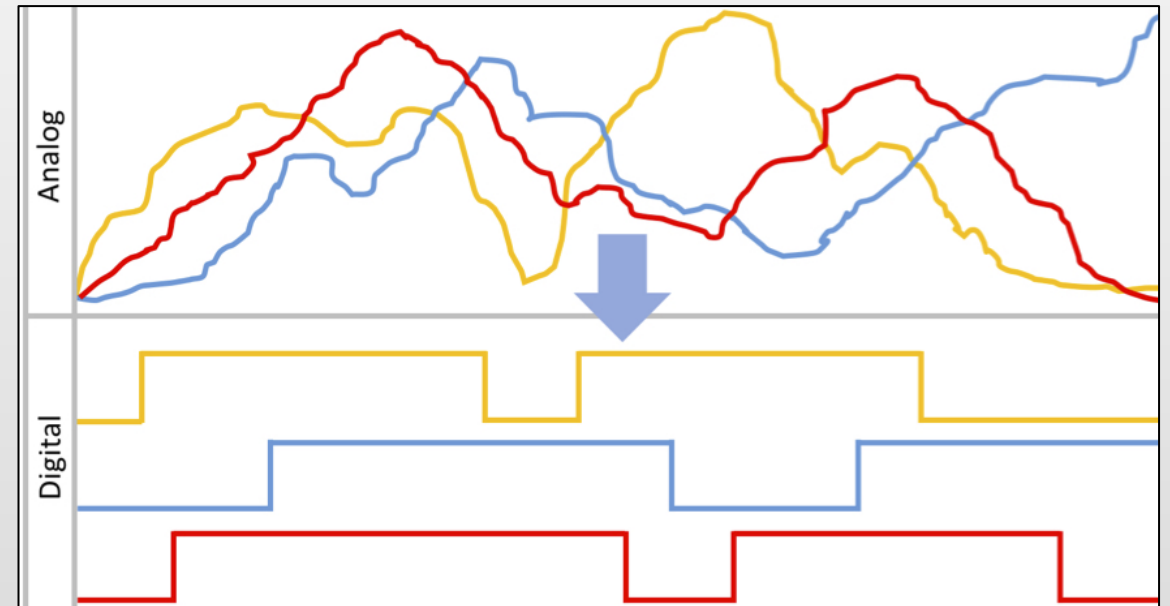
Communications





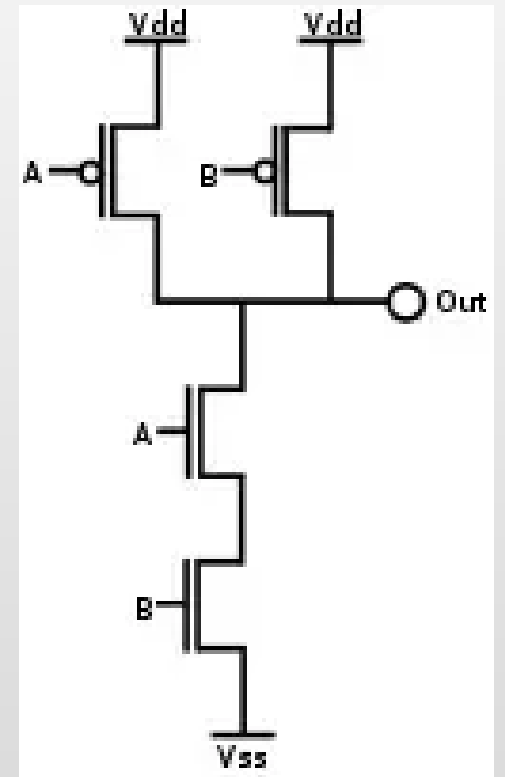
WHAT IS A DIGITAL SYSTEM?

- Discrete information is represented by physical quantities called signals, such as voltages.
- Digital systems use signals that have **two** distinct values and circuit elements that have **two** stable states.



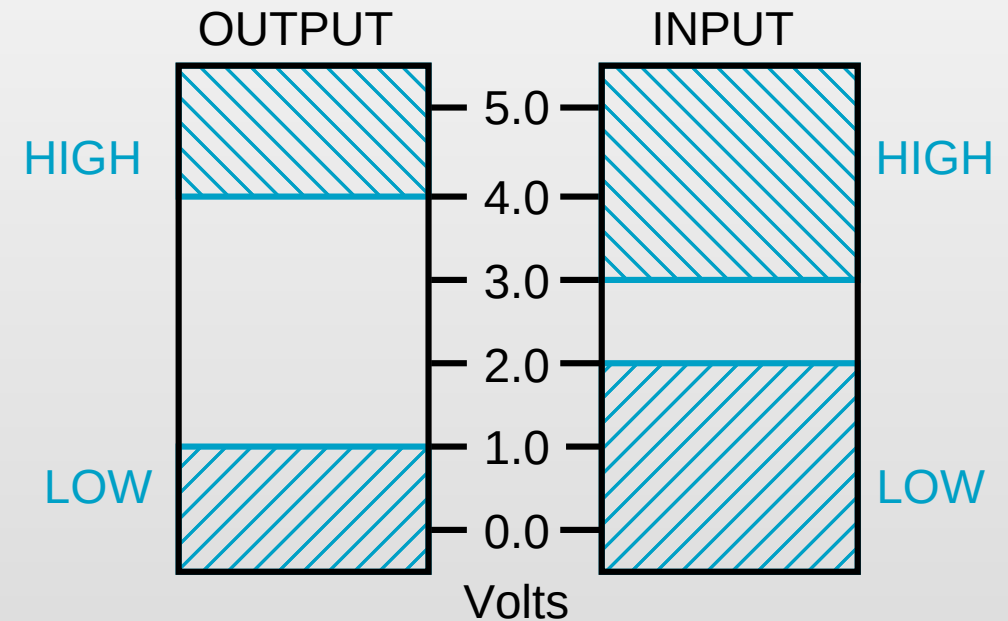
WHAT IS A LOGIC GATE?

- Electrical signals such as voltages or currents exist as analog signals having values over a given continuous range, say, 0 to 5 V, depending on whether they are on or off.
- In a digital system these voltages are interpreted to be either of two recognizable values, 0 or 1.
- Logic operations are performed by transistors in processors or FPGA's.



WHAT IS BINARY LOGIC?

- Following interchangeable terms are used to represent binary values:
 - 0 or 1
 - False (F) or true (T)
 - Low (L) or high (H)
 - On or Off
- In programming languages denoted by:
 - $\&$ for AND operation
 - $|$ for OR operation
 - $\sim$ for NOT operation





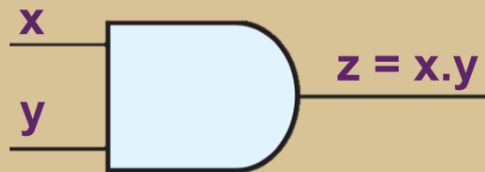
WHAT IS BINARY LOGIC?

- Binary logic deals with variables that take on two discrete values and with operations that assume logical meaning.
 - Each variable has only two distinct possible values: 1 and 0, true or false, high or low, etc.
- Binary logic consists of binary variables and a set of logical operations.
 - For example, $X \$ Y = Z$, where $\$$ indicates any logical operation.

BASIC LOGIC OPERATIONS

AND

- $x.y = z$ or $xy = z$
- $\rightarrow x$ AND y is equal to z
- $\rightarrow$ The logical operation AND is interpreted to mean that
 $z = 1$, if both $x = y = 1$
 $z = 0$, otherwise



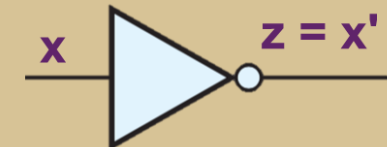
OR

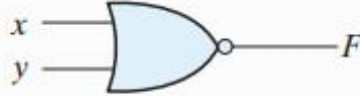
- $x+y = z$
- $\rightarrow x$ OR y is equal to z
- $\rightarrow$ The logical operation OR is interpreted to mean that
 $z = 1$, if either x or y or both = 1
 $z = 0$, otherwise



NOT

- $x' = z$ or $\bar{x} = z$
- $\rightarrow$ not x is equal to z
- $\rightarrow z = 1$, if $x = 0$ or vice versa



Name	Graphic symbol	Algebraic function	Truth table															
NAND		$F = (xy)'$	<table><tr><th>x</th><th>y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	x	y	F	0	0	1	0	1	1	1	0	1	1	1	0
x	y	F																
0	0	1																
0	1	1																
1	0	1																
1	1	0																
NOR		$F = (x + y)'$	<table><tr><th>x</th><th>y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	x	y	F	0	0	1	0	1	0	1	0	0	1	1	0
x	y	F																
0	0	1																
0	1	0																
1	0	0																
1	1	0																
Exclusive-OR (XOR)		$F = xy' + x'y$ $= x \oplus y$	<table><tr><th>x</th><th>y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	x	y	F	0	0	0	0	1	1	1	0	1	1	1	0
x	y	F																
0	0	0																
0	1	1																
1	0	1																
1	1	0																
Exclusive-NOR or equivalence		$F = xy + x'y'$ $= (x \oplus y)'$	<table><tr><th>x</th><th>y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	x	y	F	0	0	1	0	1	0	1	0	0	1	1	1
x	y	F																
0	0	1																
0	1	0																
1	0	0																
1	1	1																

BASIC LOGIC OPERATIONS

- Binary logic should not be confused with binary arithmetic.
 - **Example:** Binary arithmetic: $1 + 1 = 10$ (decimal 2), binary logic: $1 + 1 = 1$ (logic 1)
- Definitions of logical operations may be listed in a compact form called **truth tables**.

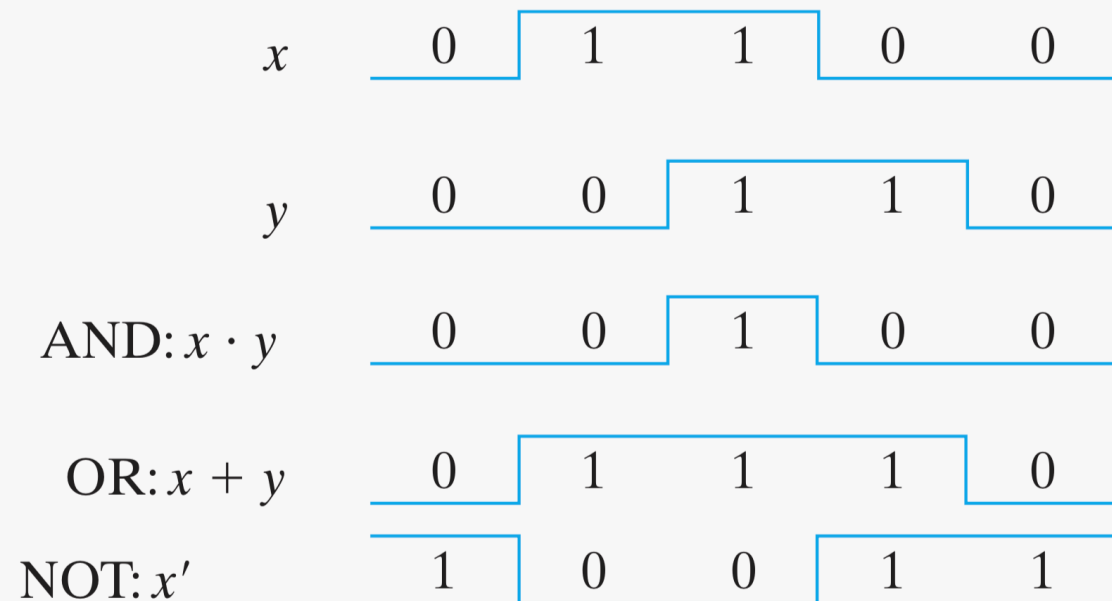
Truth Tables of Logical Operations

AND			OR			NOT	
x	y	$x \cdot y$	x	y	$x + y$	x	x'
0	0	0	0	0	0	0	1
0	1	0	0	1	1	1	0
1	0	0	1	0	1		
1	1	1	1	1	1		



BASIC LOGIC OPERATIONS

Timing Graph





LOGIC GATE APPLICATIONS

- If I have a gate that should only open when I use a voice command and biometric scan, what logic should I use?
- If I have a vending machine that asks for coins or notes, which logic should I use for currency selection?
- If I have a light that should switch on when the sunlight sensor detects no sunlight, what should be the logic?
- If I have a door sensor that only allows 1 person to cross at a time, otherwise the door remains closed, what logic may I use?
- If I have a counter which detects bit odd numbers, what logic may I use?

HOW IS DIGITAL LOGIC DEVELOPED?

Power



Manual
Fan Control



Temp
Sensor



Temp Set Control

50 °C

Digital System

Design a system which turns ON the cooler, after a delay of 1 second, when a button is pressed OR once the system reaches above the pre-defined temperature.

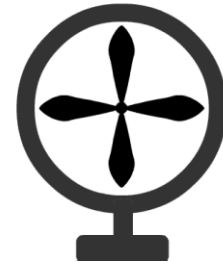
Specifications:

- Inputs: Power Button, Manual Cooler Control, Temperature Sensor, Temperature Set Control
- Output: Power Indicator, Cooling Fan
- System should work only when the system is powered ON

Power Indicator

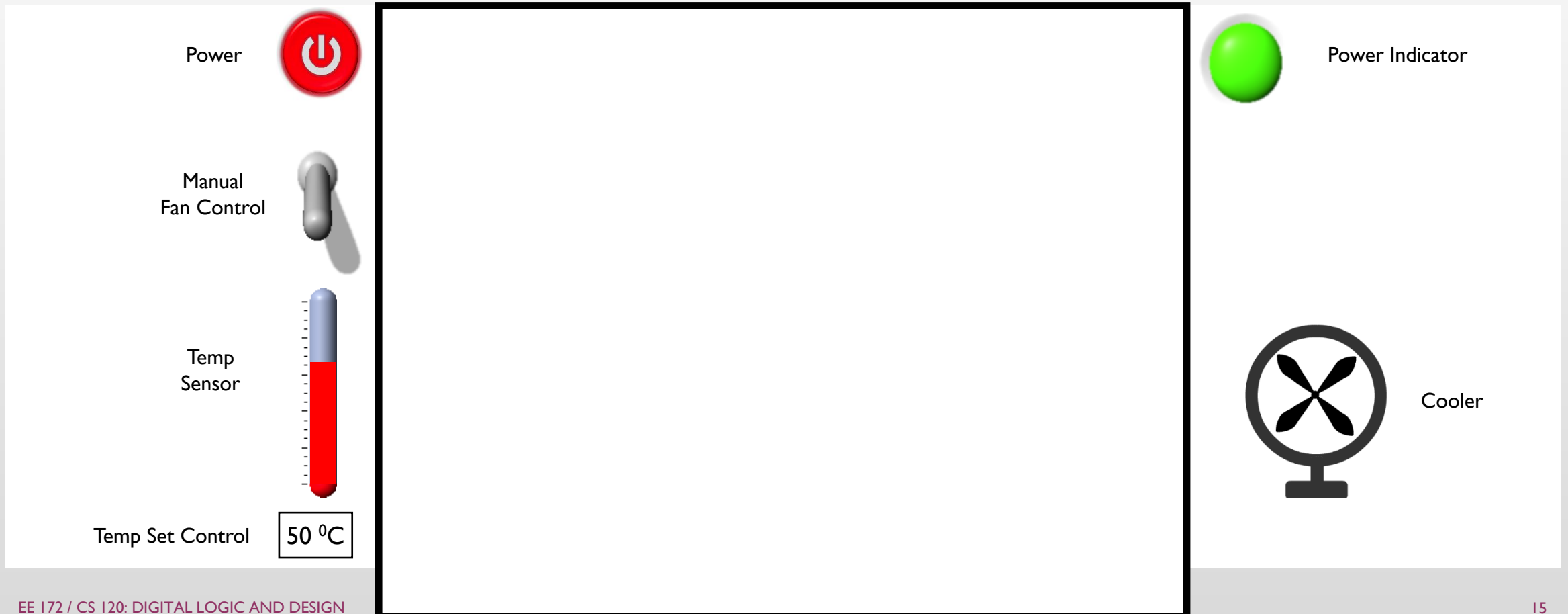


Cooler



ACTIVITY

Design a system which turns ON the cooler, after a delay of 1 second, when a button is pressed OR once the system reaches above the pre-defined temperature.

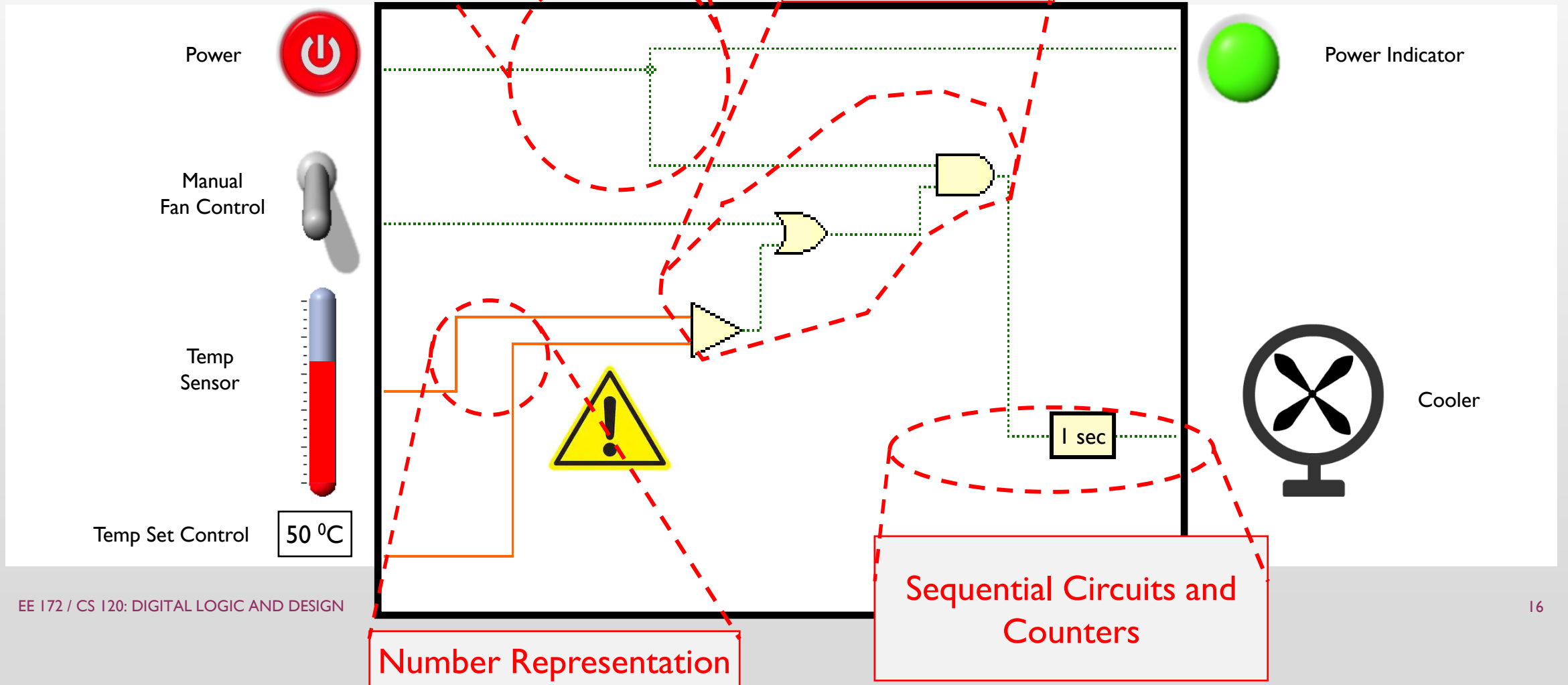


ACTIVITY



Boolean Algebra
and Laws

Logic Gates and
Combinational
Circuits



COURSE BOOK



- Digital Design with an Introduction to the Verilog HDL (5th Edition)
 - M. Morris Mano, Michael D. Ciletti

